

MGT 207 — Operations Management

Fall 2026 · UC Riverside School of Business

Danko Turcic

Associate Professor of Operations Management

Meet Your Instructor

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 Office	SBB 421
 Class Time	W 2:00 PM – 4:50 PM (AGSM 118)

Office Hours

Day	Time	Format
Wednesdays	1:00–2:00 PM	In Person (SBB 421)
Tuesdays	12:00–1:00 PM	Zoom ID: 752 601 2800

About Me

- 🏛️ **Industry** — 10 years in **Investment Banking & Commodity Trading** (Merrill Lynch, US Bank)
- 🎓 **Academia** — 10+ years at **Washington University in St. Louis (Olin School of Business) & UCR**
- 🔬 **Research** — Interface of **Operations Management & Finance · Two-Sided Markets** (so... not traditional operations management)

Course Materials

-  **Website:** <https://eLearn.ucr.edu> (homework and grades)
-  **GitHub Repo:** <https://github.com/dxt48/MGT-207> (class notes, etc.)
-  **Harvard Course Pack:** <https://hbsp.harvard.edu/import/1456740> (course pack)
-  **Helpful References:**
 - *Operations Management: Sustainability and Supply Chain Management* (any recent edition) — Heizer, Render, Munson
 - *Managerial Decision Modeling with Spreadsheets* (any edition) — Render, Stair, Balakrishnan

Course Objectives

1. Understand the **strategic role** of OM in organizations
2. Leverage OM tools for **competitive advantage** (models: inventory, capacity, queues, etc.)
3. Develop rigorous **quantitative problem-solving skills** (learn how to set up your own models in Excel, Python, etc.)

AI and Analytical Work

- **Early view (2024)**: AI like a “mediocre but not completely incompetent junior analyst”
- **Current view (2026)**: AI is “**ready for primetime**” — “saves more time than it wastes”
 - Code generation & execution
 - Running calculations & testing ideas

“AI may not be on the cusp of solving all of the world’s great problems, but chatbots are at the point where they can **collaborate** with humans opening up a different ‘way of doing analytics.’”

The Hybrid Human-AI Approach

“With difficult problems, you really want a conversation between humans and AI.”

Hybrid = Human expertise × AI power

- **AI strengths:** tireless breadth, tedious computations, rapid exploration at scale, code/visuals, quick prototyping
- **Human strengths:** deep intuition, strategic direction, spotting subtle errors, verification, creativity

Why Expertise Matters When Using AI

We often hear: *“If AI can write code and test its own solutions using loops, why do I need to learn the subject?”*

While AI can brute-force solutions and verify code computationally, **it lacks context and judgment.**

- **Formulation vs. Execution:** AI is great at solving a well-defined equation. But translating a messy, real-world business problem into the *correct* model requires domain expertise.
- **Garbage In, Garbage Out:** An AI will perfectly optimize a flawed model without questioning your faulty assumptions.
- **Evaluating Edge Cases:** A simulation loop might run successfully, but an expert knows which critical scenarios the AI forgot to test.

To Become Effective with AI

- Learn about **platforms** (GitHub, Kaggle, OpenRouter, HuggingFace, etc.)
- **Applications** (VS Code, AI Studio, NotebookLM, LM Studio, etc.)
- **Languages**: Markdown, Python
- Become effective at **prompting**
- Be critical of AI outputs which requires **domain expertise**

Grading at a Glance

Component	Weight
 Case Analysis (Simulations)	50%
 Final Case	20%
 Online Homework	30%

Three components. The bulk of your grade comes from **active participation in simulation cases** — not just a final case.

Simulation Cases

Each week features an **interactive Harvard simulation** run during class time.

- Simulations are sourced from the **Harvard Business Publishing** course pack
- You manage a real business scenario (e.g., a restaurant, supply chain, factory)
- Performance is measured on a **target metric** (most often: *total profit*)
- Conducted in teams or individually, depending on the simulation

Examples

Week	Simulation
Week 2	SimCase: Maketto
Week 3	Project Mgmt Simulation: Scope, Resources, Schedule V3

Grading Simulation Cases

Grades are based on your **percentile rank** relative to classmates on the target metric.

Percentile	Score	Grade
$\geq 95\text{th}$	100%	A+
90th – 94.9th	95%	A
85th – 89.9th	91%	A–
80th – 84.9th	88%	B+
75th – 79.9th	86%	B
$> 70\text{th} - 74.9\text{th}$ (submitted)	80%	B– (floor)
Not submitted	0%	F

Any legitimate attempt earns at least an **80% (B–)**. The lowest three case scores will be dropped.

How Case Grades Are Calculated

Step-by-step

1. **Run the simulation** during class
2. Your result (e.g., profit) is **recorded and ranked** against all classmates
3. Your **percentile rank** determines your score (see table on previous slide)
4. Scores across **all simulation cases** are averaged (lowest 3 dropped)

Key Takeaway

- **Show up and try** → guaranteed B– (80%) or better
- **Outperform classmates** → earn up to 100%
- Competition is within the class, not against a fixed curve

Final Case

- Weight: **20% of final grade**
- Format: **Take-home case analysis**
- Grading procedure is identical to the regular class cases.

There is **one final case** instead of a final exam. Steady engagement with weekly homework and cases is the best preparation.

Online Homework

- Weight: **30% of final grade**
- Administered via **Canvas**
- One assignment per topic/week

How It Works

Feature	Policy
Attempts	2 attempts per assignment
Graded attempt	Final attempt only
Collaboration	Individual work expected

Course Schedule — Fall 2026 (Part 1)

Week	Date	Topic	Simulation Case
1	Sep 30	OM for Decisions	Advertising Budget
2	Oct 07	Analytics Primer	SimCase: Maketto
3	Oct 14	Project Management	Project Mgmt Simulation: Scope, Resources, Schedule V3
4	Oct 21	Forecasting	Data Analytics Sim: Strategic Decision Making
5	Oct 28	Inventory Management	OM Simulation: Inventory Basics

Course Schedule — Fall 2026 (Part 2)

Week	Date	Topic	Simulation Case
6	Nov 04	Process & Capacity Mgmt	OM Simulation: Balancing Process Capacity
—	Nov 11	 Veterans Day — No Class	—
7	Nov 18	Waiting Lines (Queues)	OM Simulation: System Utilization
8	Nov 25	Quality Control	OM Simulation: Quality Analytics
9	Dec 02	Supply Chain Management	Global Supply Chain Management Simulation V2

Weeks 1–2: Foundations

Week 1 — OM for Decisions

- What is Operations Management?
- The role of OM in business strategy
- **Case:** Advertising Budget (linear programming intro)

Week 2 — Analytics Primer

- Decision models and quantitative tools
- Spreadsheet-based optimization
- **Sim Case:** SimCase: Maketto

Weeks 3–4: Planning & Forecasting

Week 3 — Project Management

- Work breakdown structure, Gantt charts
- Critical Path Method (CPM)
- **Sim Case:** Project Mgmt Simulation: Scope, Resources, Schedule V3

Week 4 — Forecasting

- Time series methods (moving average, exponential smoothing)
- Regression-based forecasting
- **Sim Case:** Data Analytics Sim: Strategic Decision Making

Weeks 5–6: Operations Core

Week 5 — Inventory Management

- EOQ model, safety stock, reorder points
- Trade-offs: holding vs. ordering costs
- **Sim Case:** OM Simulation: Inventory Basics

Week 6 — Process & Capacity Management

- Capacity planning and utilization
- Bottleneck analysis
- **Sim Case:** OM Simulation: Balancing Process Capacity

Weeks 7–9: Advanced Topics

Week 7 — Waiting Lines

- Queueing models: M/M/1, M/M/s
- **Sim Case:** OM Simulation: System Utilization

Week 8 — Quality Control

- Statistical process control (SPC), control charts
- **Sim Case:** OM Simulation: Quality Analytics

Week 9 — Supply Chain Management

- Global supply chain design, risk, coordination
- **Sim Case:** Global Supply Chain Management Simulation V2

How to Succeed in MGT 207

- ✓ **Attend every class** — simulations happen in real time and cannot be made up
- ✓ **Attempt homework early** — 2 attempts means room to learn, but plan ahead
- ✓ **Engage with Excel** — most quantitative problems are solved in spreadsheets
- ✓ **Ask questions** — office hours are generous; use them
- ✓ **Think strategically** — simulations reward insight, not just effort

OM is the backbone of how businesses create value. Master these tools and you'll think differently about every organization you encounter.

Course at a Glance

Instructor	Danko Turcic · dankot@ucr.edu
Course Pack	Harvard (via Canvas)
Case Analysis	50% · Percentile-ranked simulations
Final Case	20% · Take-home case analysis
Homework	30% · Canvas, 2 attempts (last graded)
Topics	OM Strategy → Project Mgmt → Forecasting → Capacity → Inventory → Queues → Quality → SCM